IBM DATA SCIENCE CAPSTONE PROJECT

Predicting Falcon 9 First-Stage Landings

From cleaned launch records to geographic insights and supervised classification

5

Published Jupyter notebooks

4

Evaluated classifiers

83.3%

Top reported test accuracy

github.com/MoOsimi/Coursera

Executive Summary

- Objective: analyze drivers of Falcon 9 first-stage landing outcomes.
- Workflow: wrangle data, explore relationships, query SQL records, map sites, and train classifiers.
- Finding: logistic regression, SVM, and KNN tie at 83.33% held-out accuracy (18 test records).
- Caution: this is a small test set; results should not be interpreted as operational reliability.

Research Question & Analytical Workflow

Can historical launch attributes help classify whether a Falcon 9 first stage lands successfully?

01 **Wrangling** → 02 **Python EDA** → 03 **SQL EDA** → 04 **Folium** → 05 **Classification**

Target: Class (1 = successful first-stage landing; 0 = otherwise).

Data Wrangling | Preparing the Target

Outcome engineering

The wrangling notebook explores categorical landing outcomes and prepares a binary Class indicator for supervised learning.

Data preparation

Launch-level fields include flight number, launch site, payload, orbit and booster attributes. The prepared dataset is reused downstream.

Repository: labs-jupyter-spacex-Data wrangling-v2.ipynb

EDA | Visual Exploration Methodology

- Scatter/categorical plots: flight number versus payload and launch site, colored by Class.
- Orbit-level success rate: mean(Class) calculated by orbit group.
- Yearly trend: average Class visualized over launch years.
- Feature engineering: one-hot encode Orbit, LaunchSite, LandingPad and Serial.

Caution: associations shown in exploratory plots do not establish causality.

EDA | Interpreting the Visual Results

Launch experience

The notebook examines how landing outcomes vary with flight number and launch location.

Orbit and mass

Payload mass is considered alongside orbit and site to reveal operating-profile differences.

Yearly success

The notebook reports an improving landing-success trend over the later observed years.

Model-ready predictors

Categorical launch attributes are converted into numeric dummy variables.

SQL EDA | Query Design

- Load SpaceX CSV into SQLite; remove records with missing dates.
- Use DISTINCT and LIKE to identify unique launch sites and filter records.
- Use SUM, AVG, MAX and nested SELECT for payload investigations.
- Use GROUP BY with COUNT(*) to rank historical landing outcomes.

SQL notebook: jupyter-labs-eda-sql-coursera_sqllite.ipynb

SQL EDA | Verified Notebook Outputs

101

Rows initially loaded to SQL

2,534.7 kg

F9 v1.1 mean payload

4

Distinct launch site labels

Historical outcomes | 2010-06-04 through 2017-03-20

No attempt: 10 • Drone-ship successes: 5 • Drone-ship failures: 5
Ground-pad successes: 3 • Controlled ocean: 3

Quality note: the NASA payload query filters %NASA% rather than NASA (CRS) exclusively.

Folium | Spatial Analytics Methodology

Geographic markers

Launch sites represented with latitude and longitude, circles and popup labels.

Outcome clusters

MarkerCluster organizes records; green/red marker coding corresponds to Class 1/0.

Proximity analysis

MousePosition and Haversine distance support coastline/transport proximity investigation.

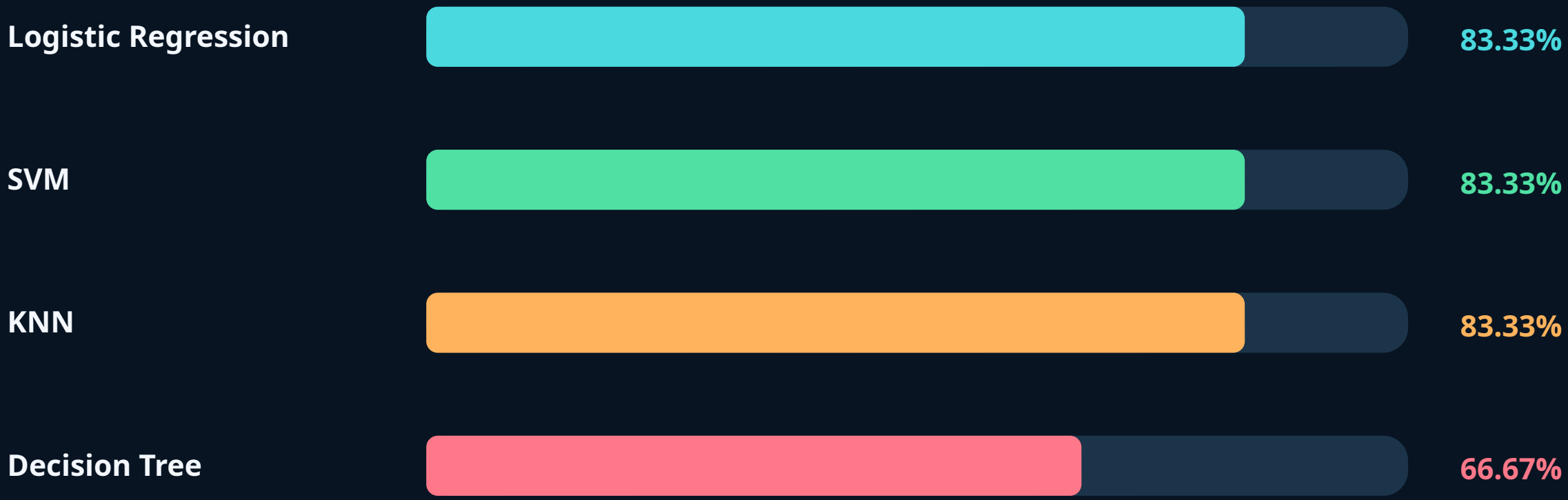
Implementation status: repository notebook still contains unfinished TODO cells for final proximity markers and lines.

Classification | Evaluation Design

- Predictors: numeric launch attributes plus one-hot encoded categories.
- StandardScaler applied to feature matrix in the saved notebook.
- 80/20 training/test split with random_state = 2; test set size = 18.
- 10-fold GridSearchCV used for hyperparameter selection.
- Compared logistic regression, SVM, decision tree and KNN.

Technical caveat: scaling occurs before the train/test split in the notebook; a Pipeline would reduce leakage risk.

Model Comparison | Held-out Accuracy



Three classifiers achieved 15/18 correct predictions; the decision tree achieved 12/18.

Model Selection | What the Scores Mean

82.14%

Logistic regression CV accuracy

84.82%

SVM CV accuracy

85.71%

Decision tree CV accuracy

- Cross-validation and test rankings differ: the decision tree has the highest CV result but the lowest held-out result.
- A tie in test accuracy does not guarantee equivalent precision, recall, calibration, or operational utility.
- Only 18 held-out observations: a single additional correct prediction changes test accuracy by 5.56 percentage points.

Conclusions & Recommendations

- Prepared launch records support a reproducible, end-to-end classification exercise.
- Orbit, payload, site, experience and reuse attributes enable useful exploratory comparisons.
- Logistic regression, SVM and KNN tie at 83.33% accuracy on this test split.
- Improve rigor with leakage-safe preprocessing, stratified repeated validation and richer metrics.
- Complete and validate Folium map annotations before presenting precise distance claims.

Evidence & Repository Links

Data wrangling	labs-jupyter-spacex-Data wrangling-v2.ipynb
Python EDA	jupyter-labs-eda-dataviz-v2.ipynb
SQL EDA	jupyter-labs-eda-sql-coursera_sqllite.ipynb
Folium	lab-jupyter-launch-site-location-v2.ipynb
Classification	SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Full repository: github.com/MoOsimi/Coursera